

Project Report

1. INTRODUCTION:

1.1 Project Overview-

Heritage Treasures is a data-driven analytical dashboard project developed to explore and visualize UNESCO World Heritage Sites across the globe. The project focuses on analysing heritage sites based on categories (Cultural, Natural, Mixed), geographical regions, area coverage, year of inscription, and danger status.

Using data visualization and analytical techniques, the dashboard provides meaningful insights such as:

- Top 10 regions by total heritage area
- Category-wise distribution of heritage sites
- Site count per region
- Analysis of heritage sites prone to extinction
- Year-wise trend and forecasting of heritage site growth
- Exportable analytical reports

The system integrates data preprocessing, visualization, forecasting, and reporting into a unified dashboard platform. It enables users to interactively explore global heritage data and identify patterns, risks, and growth trends over time.

1.2 Purpose-

The purpose of the Heritage Treasures project is to:

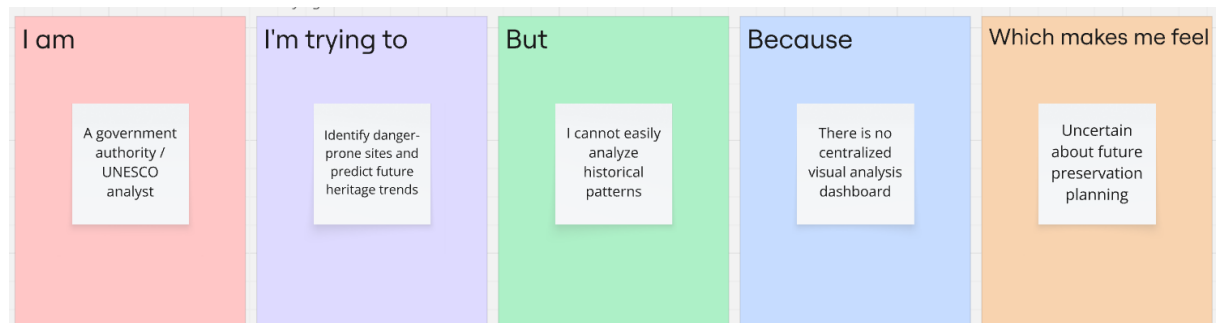
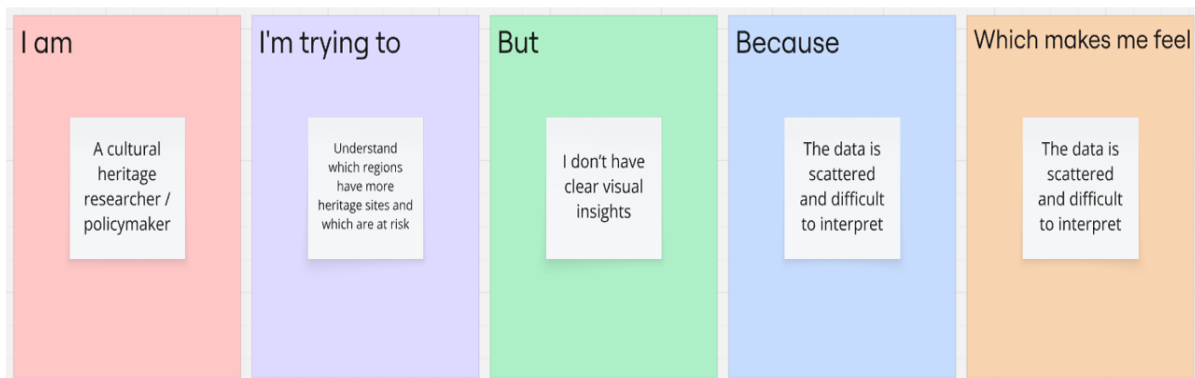
1. Analyze and visualize global UNESCO heritage data in an interactive and user-friendly format.
2. Identify regions with the highest heritage concentration and area coverage.
3. Detect heritage sites that are in danger and require conservation focus.
4. Forecast future trends in heritage site inscription using historical data.
5. Support decision-making for researchers, policymakers, and heritage conservation organizations.

6. Provide a scalable analytical dashboard that transforms raw heritage data into actionable insights.

This project aims to promote awareness about global heritage preservation while demonstrating practical implementation of data analytics, dashboard development, and predictive modeling techniques.

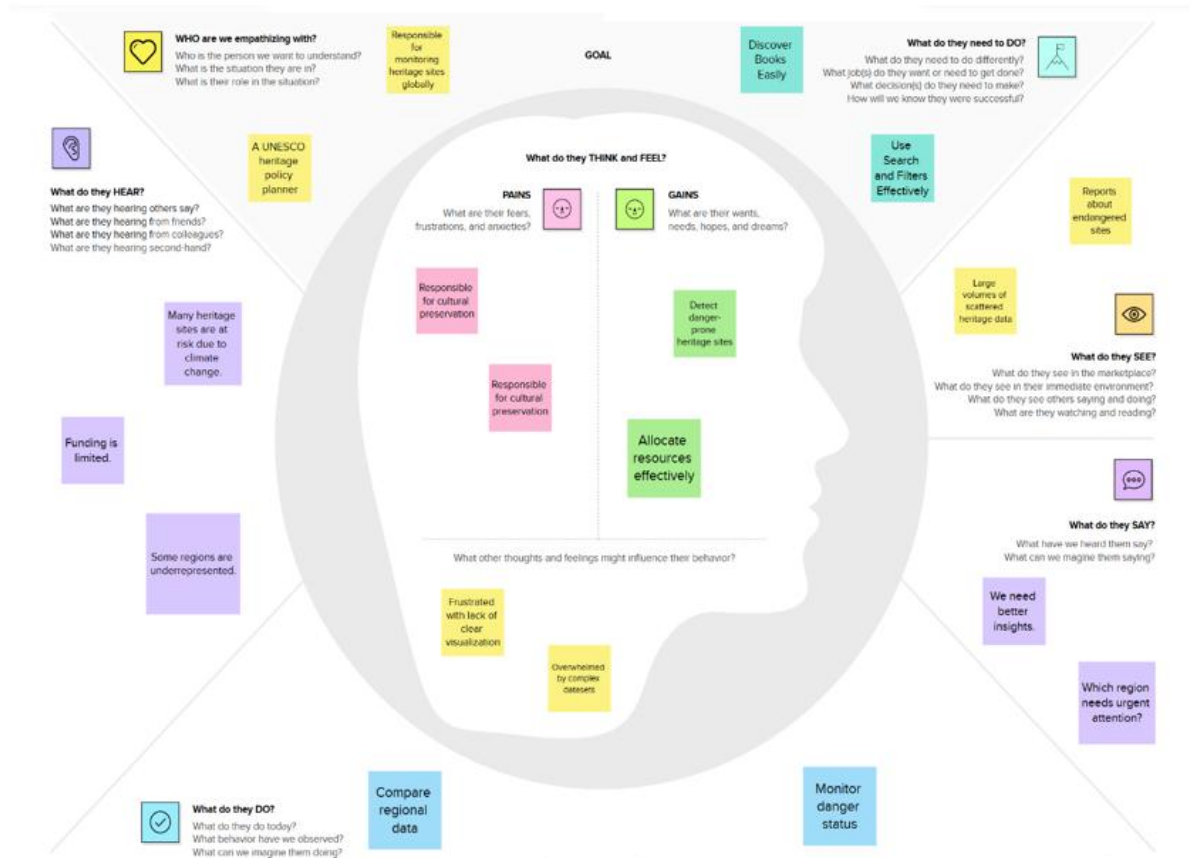
2. IDEATIONPHASE:

2.1 Problem Statement-



Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A cultural heritage researcher / policymaker	Understand which regions have more heritage sites and which are at risk	I don't have clear visual insights	The data is scattered and difficult to interpret	Confused and unable to make informed decisions
PS-2	A government authority / UNESCO analyst	Identify danger-prone sites and predict future heritage trends	I cannot easily analyze historical patterns	There is no centralized visual analysis dashboard	Uncertain about future preservation planning

2.2 Empathy Map Canvas-



2.3 Brainstorming-

1

Define your problem statement

What problem are you trying to solve? Frame your problem as a How Might We statement. This will be the focus of your brainstorm.

⌚ 5 minutes

PROBLEM

How might we help users easily discover, explore, and understand global heritage sites through an interactive digital platform while promoting cultural awareness and preservation?

Key rules of brainstorming

To run a smooth and productive session

- Stay in topic.
- Defer judgment.
- Go for volume.
- Encourage wild ideas.
- Listen to others.
- If possible, be visual.

2

Brainstorm

Write down any ideas that come to mind that address your problem statement.

🕒 10 minutes

NITHISH

Which regions have the highest number of heritage sites?

What factors make sites become "In Danger"?

Is the number of endangered sites increasing over years?

Are natural sites more at risk than cultural sites?

Which countries contribute most heritage area?

Are large-area sites more likely to be endangered?

Which continent needs more conservation focus?

Can tourism impact heritage risk?

How has inscription trend changed over decades?

3

Group ideas

Take turns sharing your ideas while clustering similar or related notes as you go. Once all sticky notes have been grouped, give each cluster a sentence-like label. If a cluster is bigger than six sticky notes, try and see if you can break it up into smaller sub-groups.

🕒 20 minutes

Regional Distribution

Regions with highest heritage sites

Site count per region

Continent-wise comparison

Map-based visualization

Category-Based Insights

Cultural vs Natural vs Mixed

Which category dominates?

Category vs danger status

Category growth over years

Endangered / Risk Analysis

Sites marked "In Danger"

Increase/decrease over years

Countries with most endangered sites

Impact of conflict or climate

Time Trend & Forecasting

Year-wise inscription trend

Growth pattern over decades

Future forecasting

Trend prediction

Area & Impact Analysis

Largest heritage areas

Area vs risk correlation

Top 10 regions by area

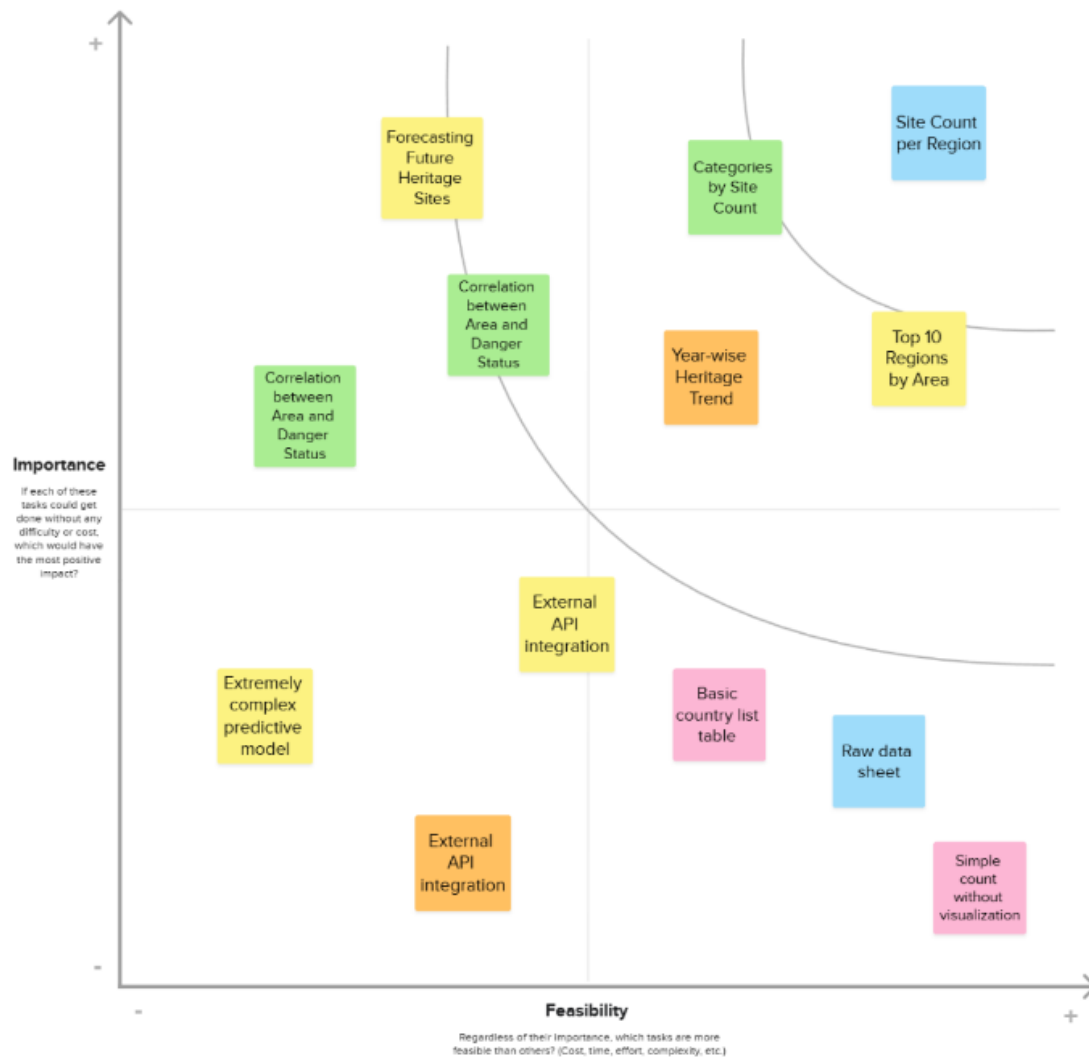
Site density

4

Prioritize

Your team should all be on the same page about what's important moving forward. Place your ideas on this grid to determine which ideas are important and which are feasible.

🕒 20 minutes



3. REQUIREMENT ANALYSIS:

3.1 Customer Journey map-

Scenario: [Existing experience through a product or service]	Entice How does someone become aware of this service?	Enter What do people experience as they begin the process?	Engage In the core elements in the process, what happens?	Exit What do people typically experience as the process finishes?	Extend What happens after the experience is over?
Experience steps What does the person (or persona) at the center of this scenario typically experience in each step?	Finds project through college/demoSees dashboard previewInterested in heritage data insights	Open dashboardView main homepage	Analyze top 10 regionsCheck danger sitesExplore year forecastingCompare categoriesExport insights	Download reportTake screenshot	Share with classmatesUse in presentationSuggest improvements
Interactions What interactions do they have at each step along the way? • People: Who do they see or talk to? • Places: Where are they? • Things: What digital touchpoints or physical objects do they use?	InstructorProject presentationWeb Demo Link	Dashboard interfaceFilters (Region, Category, Danger Status)	Charts (Map, Bar, Line, Pie)Hover tooltipsExport optionsDrill-down filteringToggle between maps and charts	Export to PDF/PNGSave filtered results	Presentation slidesFeedback discussionFuture enhancement brainstorming
Goals & motivations At each step, what is a person's primary goal or motivation? ("Help me..." or "Help me avoid...")	Help me explore heritage data easilyHelp me understand global heritage trendsHelp me find relevant heritage statistics	Filters (Region, Category, Danger Status)Help me see country-level data	Help me identify trendsHelp me compare regionsHelp me find endangered sitesHelp me detect patterns over timeHelp me which regions are at risk	Help me present findingsHelp me share insights	Help me improve analysisHelp me apply to real world dataHelp me conduct deep research
Positive moments What steps does a typical person find enjoyable, productive, fun, motivating, delightful, or exciting?	Clean UI designClean UI designHelp me find relevant heritage statistics	Interactive filtersMap visualization loads properly	Forecasting graphTreemap for danger sitesInteractive map coloringClear regional comparisonSmooth transitions when data updates instantly	Clean exportClear charts	Good academic feedbackVisual appeal appreciatedSees future research potential
Negative moments What steps does a typical person find frustrating, confusing, engaging, costly, or time-consuming?	Not sure what insights are availableAdd project description pop-upAdd project description pop-up	Too many filters at onceConfusing Terminology	Too many visuals on one screenHard to interpret large numbersData misrepresentation risksLimited explanation for complex metricsData misrepresentation risks	Manual export processNo auto-generated summary	Limited real time dataStatic datasetNo advanced analytics
Areas of opportunity How might we make each step better? What ideas do we have? What have others suggested?	Add intro screenAdd project description pop-upAdd sample insights preview	Add filter tooltipsAdd reset button	Add dashboard summary cardsAdd KPI indicatorsImprove color legend clarityAdd insights explanation panelAdd comparison mode	One click report exportAutomated summary generation	Add live UNESCO APIAdd predictive AI modelAdd user comparison mode

3.2 Solution Requirement-

Functional Requirements:

Following are the functional requirements of the proposed solution.

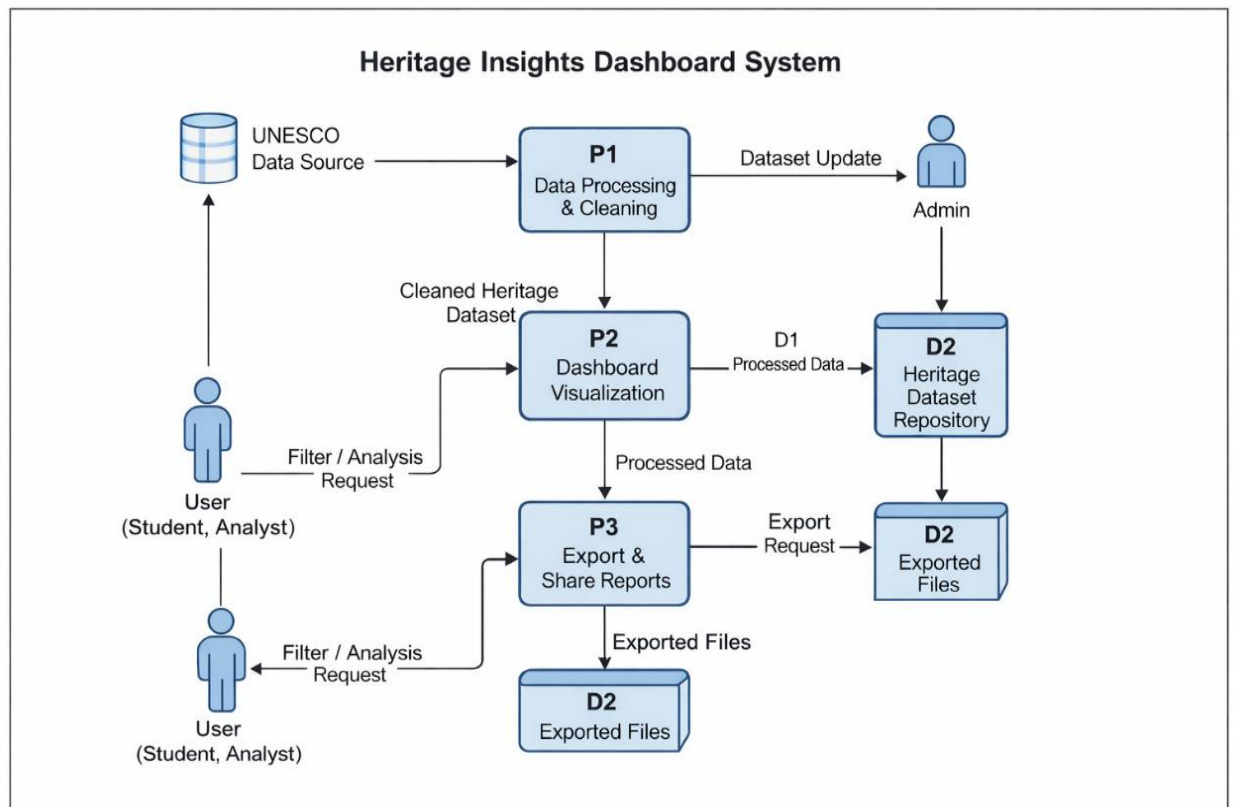
FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Visualization	Display interactive world map of heritage sites
		Show Top 10 Regions by Area
		Show Categories (Cultural, Natural, Mixed) distribution
FR-2	Data Filtering & Analysis	Filter by Region
		Filter by Year Inscribed
		Filter by Danger Status
FR-3	Comparative Analysis	Compare regions by site count
		Compare area coverage by country
		Analyse trend of heritage inscriptions over years
FR-4	Forecasting & Trend Analysis	Display year-wise trend line chart
		Forecast future heritage counts
FR-5	Data Export	Export dashboard as PDF/Image
		Allow download of summarized data
FR-6	Danger Site Analysis	Show endangered sites distribution
		Analyse danger sites vs area coverage

Non-functional Requirements:

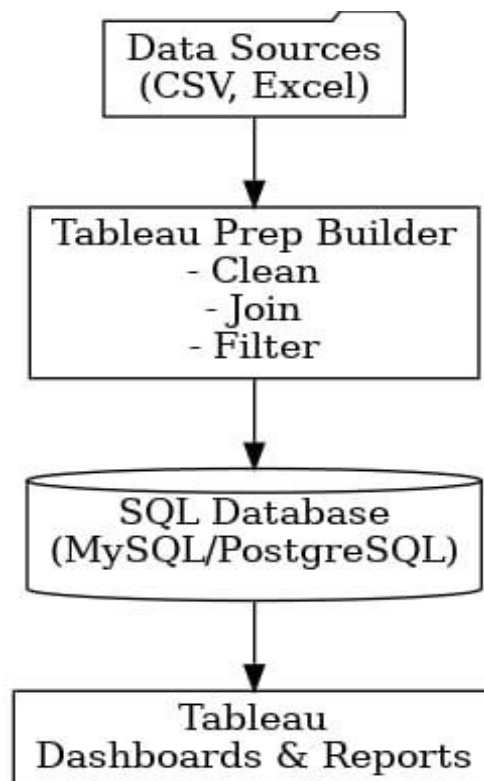
Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Dashboard must be simple, intuitive, and easy to navigate with clearly labelled charts.
NFR-2	Security	Data integrity must be maintained; no unauthorized data modification.
NFR-3	Reliability	Dashboard should display accurate visualizations without errors or crashes.
NFR-4	Performance	Visualizations must load within 3–5 seconds even with filters applied.
NFR-5	Availability	Dashboard should be accessible whenever the system is running (local/server environment).
NFR-6	Scalability	System should support additional datasets and future feature expansion (e.g., ML prediction).
NFR-7	Maintainability	Code and dashboard design should be modular and easy to update.

3.3 Data Flow Diagram-



3.4 Technology Stack-



4. PROJECT DESIGN :

4.1 Problem Solution Fit-

Problem-Solution fit canvas 2.0

Purpose / Vision

Define CS, fit into	1. CUSTOMER SEGMENT(S) CS Who is your customer? - History enthusiasts - Students & researchers - Tourists (domestic & international) - Cultural explorers - Museums / educational institutions	6. CUSTOMER CC What constraints prevent your customers from taking action or limit their choices of solutions? - Lack of structured information - Time constraints, Budget concerns - Language barriers, No single trusted platform	5. AVAILABLE SOLUTIONS AS Which solutions are available to the customers when they face the problem or need to get the job done? What have they tried in the past? What pros & cons do these solutions have? - WGoogle search - Wikipedia - Travel blogs - YouTube videos, Tourism websites	Explore AS,
	2. JOBS-TO-BE-DONE / PROBLEMS J&P Which jobs-to-be-done (or problems) do you address for your customers? There could be more than one; explore different sides. - Discover authentic information about heritage sites - Learn historical significance easily - Plan cultural visits - Access reliable, organized heritage data - Explore hidden or lesser-known heritage spots	9. PROBLEM ROOT CAUSE RC What is the real reason that this problem exists? What is the back story behind the need to do this job? - Heritage information is not centralized - Lack of interactive digital platforms - Limited awareness of lesser-known heritage spots - Poor digital presentation of cultural data	7. BEHAVIOUR BE What does your customer do to address the problem and get the job done? i.e. directly related: find the right solar panel installer, calculate usage and benefits; indirectly associated: customers spend free time on volunteering work (i.e. Greenpeace) - Search online - Watch travel videos - Read blogs - Ask friends - Visit tourist offices	
Identify strong TR & EM	3. TRIGGERS TR What triggers customers to act? - Planning a trip, Academic project or research - Cultural event or festival, Social media exposure - Curiosity after hearing about a monument	10. YOUR SOLUTION SL Househunt Web App - Centralized heritage database - Interactive maps - Verified historical content - User-friendly interface - Cultural Storytelling approach - Educational support	8. CHANNELS of BEHAVIOUR CH 8.1 ONLINE What kind of actions do customers take online? Extract online channels from #7 Google Search, Instagram, YouTube, Travel forums, Tourism websites 8.2 OFFLINE What kind of actions do customers take offline? Extract offline channels from #7 and use them for customer development. Travel agencies, Local guide, books, tourism offices	Extract online & offline CH of BE
	4. EMOTIONS: BEFORE / AFTER EM How do customers feel when they face a problem or a job and afterwards? - Before: Confused, stressed, insecure, overwhelmed. - After: Confident, relieved, in control, excited to move.			

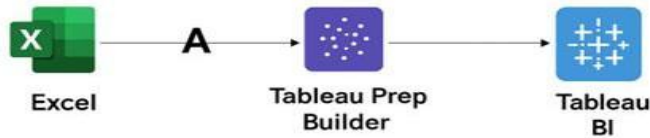
4.2 Proposed Solution-

Proposed Solution Template:

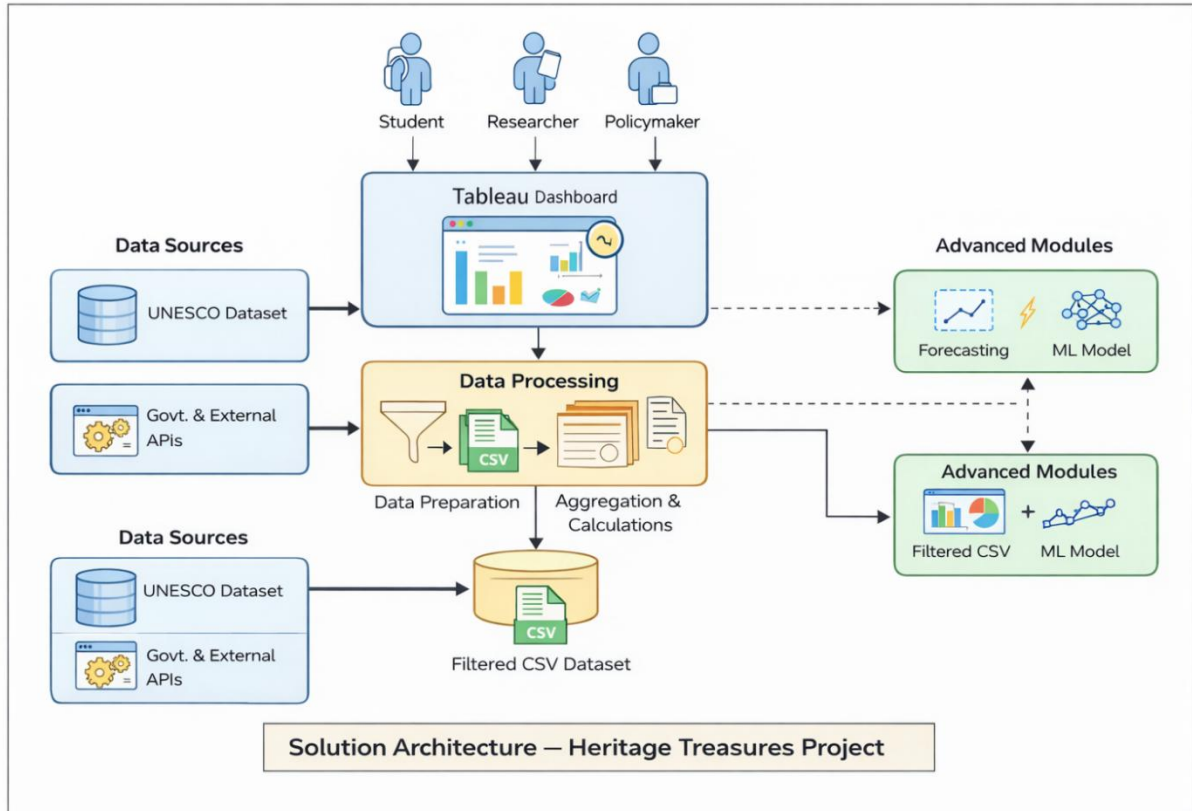
Project team shall fill the following information in the proposed solution template.

S. No	Parameter	Description
1	Problem Statement (Problem to be solved)	UNESCO heritage site data is large and scattered across multiple categories, regions, and years. It is difficult to quickly analyze trends such as endangered sites, regional distribution, category distribution, and area coverage. Decision-makers lack a centralized, visual, and interactive platform to derive meaningful insights.
2	Idea / Solution Description	Develop an interactive Heritage Dashboard that visually analyzes UNESCO heritage data using maps, treemaps, forecasting models, and region/category comparisons. The dashboard enables users to explore site counts, endangered sites, area distribution, and year-wise trends in a simple and intuitive manner.
3	Novelty / Uniqueness	The solution integrates multiple analytical perspectives (Geographical Analysis, Forecasting, Danger Status, Category Comparison, Area Analysis) into one unified dashboard. It combines visual storytelling with predictive analysis, making complex heritage data easily understandable.
4	Social Impact / Customer Satisfaction	Helps researchers, policymakers, and students understand heritage preservation trends. Raises awareness about endangered heritage sites and supports informed decision-making. Enhances educational understanding of global cultural and natural heritage distribution.
5	Business Model (Revenue Model)	The dashboard can be monetized through premium analytical reports, subscription-based access for research institutions, data-driven consulting services, or licensing to government and tourism departments. Sponsored insights and heritage analytics reports can also generate revenue.
6	Scalability of the Solution	The solution is scalable as it can integrate additional datasets (tourism revenue, environmental impact, climate data). It can be deployed on cloud platforms, expanded into a full web analytics portal, or enhanced with AI-driven predictive analytics for global heritage monitoring.

4.3 Solution Architecture-1).



2).



5. PROJECT PLANNING & SCHEDULING:

5.1 Project Planning-

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Data Collection & Cleaning	USN-1	As a system, I can import UNESCO heritage dataset for analysis	2	High	Nithish
Sprint-1	Data Collection & Cleaning	USN-2	As a system, I can clean null and inconsistent values in dataset	3	High	Nithish
Sprint-1	Data Preparation	USN-3	As a system, I can filter and prepare dataset for visualization	2	High	Nithish
Sprint-1	Dashboard Development	USN-4	As a user, I can view Top 10 Regions by Area visualization	3	High	Nithish
Sprint-1	Dashboard Development	USN-5	As a user, I can view Categories by Site Count	2	High	Nithish
Sprint-2	Advanced Analysis	USN-6	As a user, I can analyze danger sites prone to extinction	3	High	Nithish
Sprint-2	Advanced Analysis	USN-7	As a user, I can analyze site count per region	2	Medium	Nithish
Sprint-2	Forecasting	USN-8	As a user, I can view year-wise forecasting of heritage sites	3	High	Nithish
Sprint-2	Machine Learning	USN-9	As a system, I can predict future heritage trends using ML model	4	Medium	Nithish
Sprint-2	Reporting	USN-10	As a user, I can export dashboard reports	2	Medium	Nithish

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed	Sprint Release Date (Actual)
Sprint-1	12	14 Days	01 Jan 2026	14 Jan 2026	12	14 Jan 2026
Sprint-2	14	14 Days	15 Jan 2025	28 Jan 2026	14	28 Jan 2026

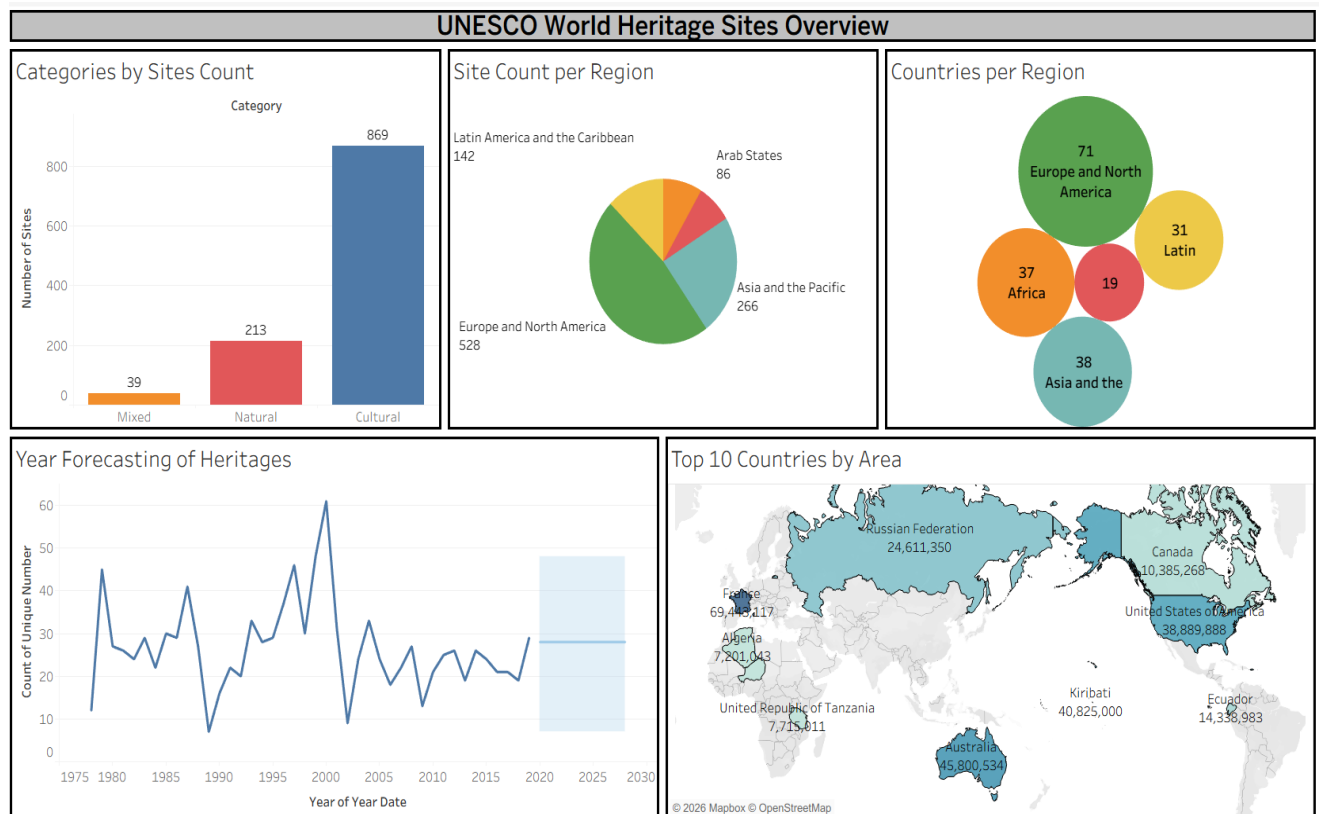
6. FUNCTIONAL AND PERFORMANCE TESTING:

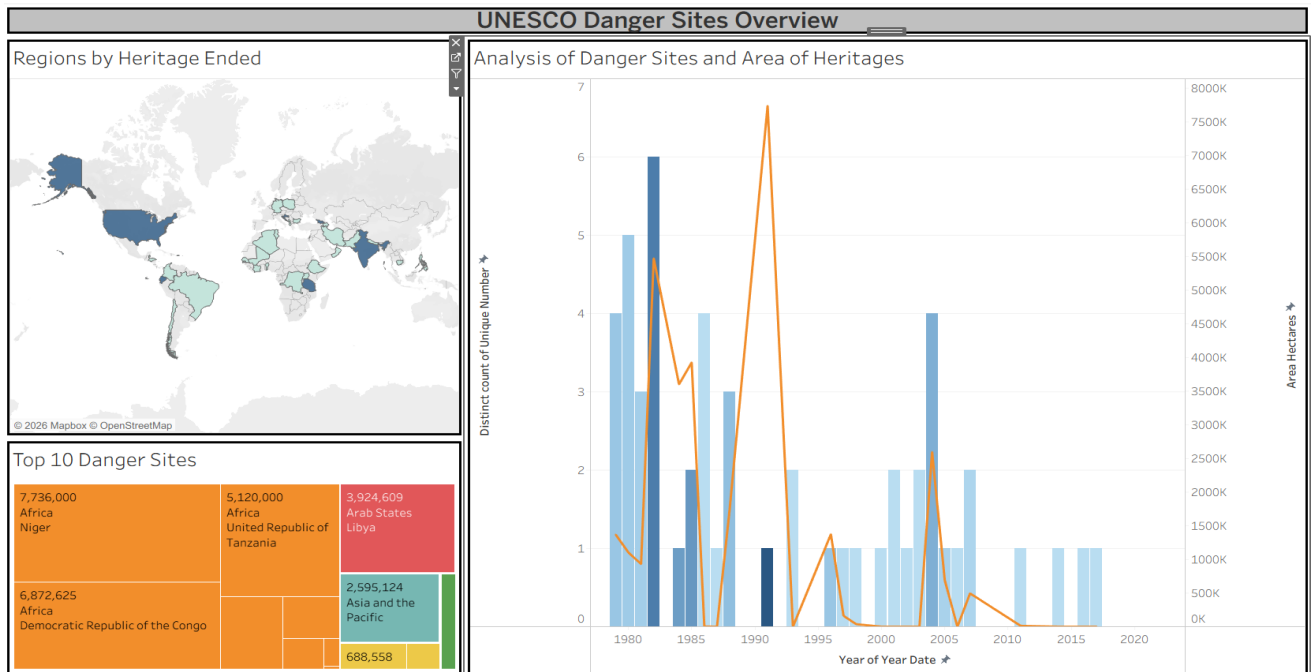
6.1 Performance Testing-

S.No.	Parameter	Screenshot / Values
1	Data Rendered	UNESCO Heritage Dataset (Heritage Name, Category, Region, Country, Year Inscribed, Area in Hectares, Danger Status, Latitude, Longitude). Total records rendered: ~1000+ sites (filtered dataset used in Tableau).
2	Data Preprocessing	Removed null values, cleaned inconsistent region names, standardized date formats, converted area hectares to numeric, created calculated fields for site count & forecasting, grouped regions.
3	Utilization of Filters	Filters used: Region filter, Danger Status filter, Year filter, Category filter, Country filter. Dynamic filtering applied in dashboard for interactive analysis.
4	Calculation Fields Used	CNTD(Name) – Site Count, <u>SUM(area hectares)</u> – Total Area, Forecast calculation (Tableau Forecast), % of Total Sites, Year-wise trend calculation, Danger site count calculation.
5	Dashboard Design	No. of Visualizations / Graphs – 2 (UNESCO World Heritage Sites Overview, UNESCO Danger Sites Overview).
6	Story Design	No. of Visualizations / Graphs – 8 (Data Overview, Regional Analysis, Category Insights, Danger Analysis, Forecasting Trends, Summary Insights).

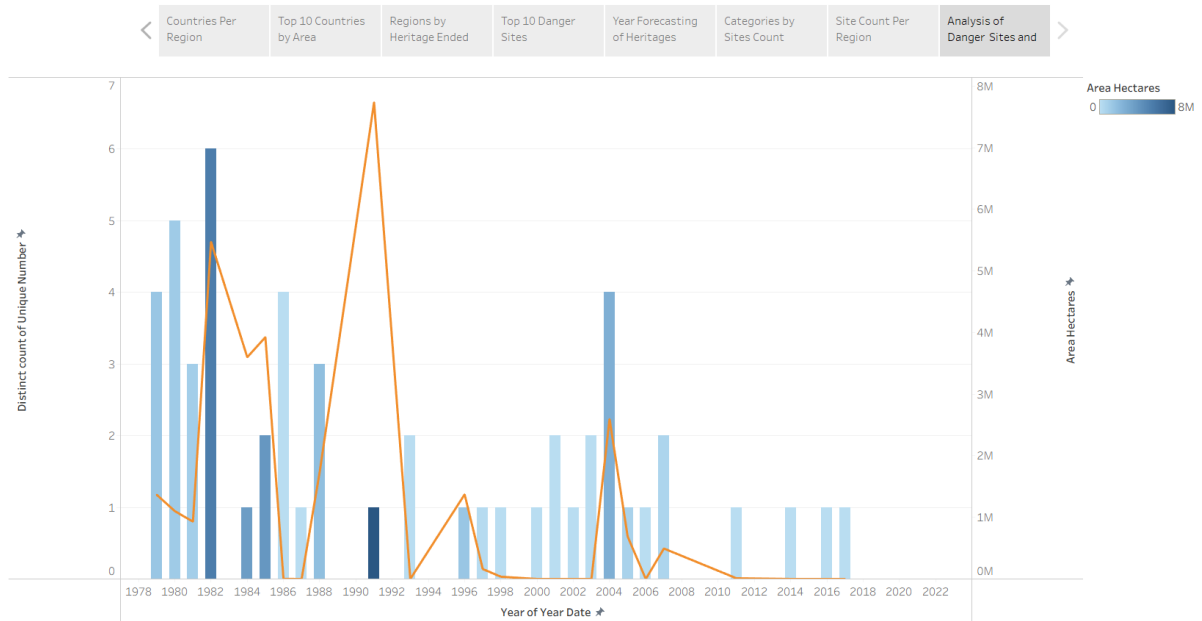
7. RESULTS:

7.1 Output Screenshots-





Overview on UNESCO Heritage Sites



8. ADVANTAGES & DISADVANTAGES:

Advantages:

1. Data-Driven Insights

The dashboard converts raw UNESCO heritage data into meaningful visual insights, helping users understand trends, distributions, and patterns easily.

2. Interactive Visualization

Users can apply filters (region, category, year, danger status) to dynamically explore the data and generate customized insights.

3. Improved Decision Support

- The project helps researchers, policymakers, and conservation authorities identify:
- Regions with high heritage density
- Sites prone to extinction
- Growth trends over years

4. Forecasting Capability

The inclusion of year-wise forecasting allows prediction of future heritage site trends, supporting strategic planning and conservation efforts.

5. Centralized Dashboard

All key analytics (area analysis, category distribution, regional analysis, danger sites, forecasting) are available in one integrated dashboard.

6. Exportable Reports

Users can export dashboard results for presentations, documentation, or further research.

7. Scalable Architecture

- The system can be extended with:
- Real-time data updates
- Machine learning improvements
- Integration with external APIs

Disadvantages:

1. Dependency on Dataset Accuracy

The insights depend entirely on the quality and completeness of the UNESCO dataset. Missing or outdated data can affect results.

2. Limited Real-Time Updates

If the dataset is static, the dashboard may not reflect newly added heritage sites immediately.

3. Forecasting Limitations

Predictions are based on historical trends and may not account for:

- Political decisions
- Natural disasters
- Policy changes
- Sudden heritage recognition growth

4. No Real-Time Monitoring

The system analyzes historical data but does not provide live monitoring of endangered sites.

5. Technical Complexity

Developing forecasting models, data preprocessing, and dashboard optimization requires technical expertise.

6. Performance Constraints

With very large datasets, dashboard performance may slow down if not optimized properly.

9. CONCLUSION:

The Heritage Treasures project successfully demonstrates how data analytics and visualization techniques can transform raw UNESCO World Heritage Site data into meaningful insights. By integrating data preprocessing, interactive dashboards, regional comparisons, danger analysis, and forecasting models, the system provides a comprehensive analytical platform for understanding global heritage distribution and trends.

The project enables users to explore heritage data through dynamic filters and visual representations such as maps, treemaps, bar charts, pie charts, and line graphs. It highlights key insights such as the regions with the highest number of heritage sites, sites prone to extinction, category distribution, and historical growth trends. The forecasting feature further enhances the analytical depth by predicting future heritage site trends based on historical data.

Overall, the project promotes awareness of global heritage conservation while demonstrating practical implementation of data visualization, analytics, and predictive modeling. It provides a scalable foundation that can be extended with real-time data integration, advanced machine learning models, and enhanced reporting features in the future.

The Heritage Treasures dashboard stands as an effective analytical tool that supports research, policy-making, and educational purposes, contributing toward better understanding and preservation of world heritage sites.

10. FUTURE SCOPE:

The Heritage Treasures project can be further enhanced by integrating real-time UNESCO data updates and incorporating advanced machine learning models for more accurate forecasting. Future improvements may include adding environmental and climate risk analysis to assess threats to heritage sites, developing a mobile-friendly version for wider accessibility, and deploying the system on a scalable cloud platform. Additionally,

integrating tourism and economic impact data could provide deeper insights for policymakers and conservation authorities, making the dashboard a more comprehensive decision-support system.

11. APPENDIX:

Dataset Link- <https://www.kaggle.com/datasets/ujwalkandi/unesco-world-heritage-sites/data?select=whc-sites-2019.csv>

Project Demo Link-

<https://drive.google.com/file/d/1YNhLkICig25MNTMTrIJqF1BdzQizuZ37/view?usp=sharing>

GitHub- <https://github.com/nithishattili/Heritage-Treasures>